

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/01

Paper 1 Multiple Choice

24 Sept 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **17** printed pages and **1** blank page.



Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 *Use of the Data Booklet is relevant to this question.*

Which particle has more protons than electrons **and** neutrons? ($D = {}^2_1\text{H}$)

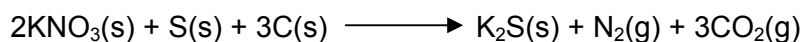
- A** H_3O^+ **B** D_3O^+ **C** NH_2^- **D** OD^-

- 2 When potassium chlorate(V) KClO_3 , is heated at its melting point, it disproportionates to potassium chlorate(VII) KClO_4 , and potassium chloride.

What is the maximum number of moles of potassium chlorate(VII) which could be produced from 0.1 mol of potassium chlorate(V)?

- A** 0.1 **B** 0.08 **C** 0.075 **D** 0.06

- 3 In the middle of the 15th century, 'black powder' was widely used by the Arabs as propellant for muzzle-loading cannons. 'Black powder' consisted of a mixture of potassium nitrate, sulfur and charcoal (carbon) in the molar ratio shown in the equation below.



Given that 2.00 mol of gas could just expel the cannonball out of the cannon tube, what is the minimum mass of 'black powder' (in g), required for this to happen?

- A** 101 **B** 136 **C** 202 **D** 270

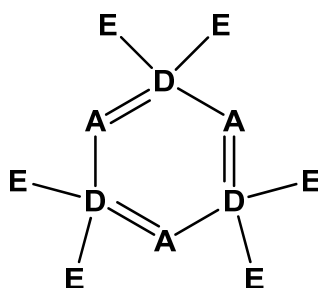
- 4 Which element has an equal number of paired and unpaired electrons in its orbitals with principal quantum number 2?

- A** beryllium
B carbon
C nitrogen
D oxygen

- 5 The boiling point of water is greater than that of hydrogen fluoride, HF.

Which statement is a correct explanation of this?

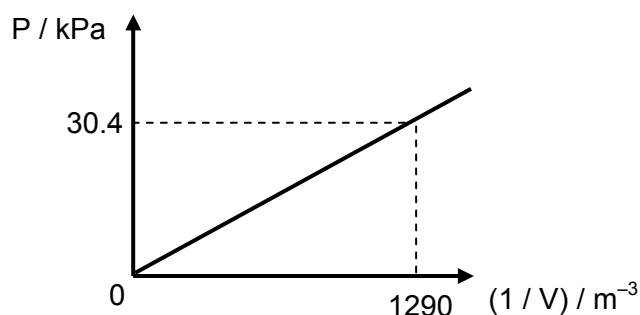
- A** Each hydrogen bond formed between water molecules is stronger than that formed between hydrogen fluoride molecules.
- B** There are more atoms in a water molecule than there are in hydrogen fluoride molecule, so van der Waals' forces are stronger in water.
- C** There are, on average, more hydrogen bonds between water molecules than there are between hydrogen fluoride molecules.
- D** The water molecule has a greater permanent dipole than the hydrogen fluoride molecule.
- 6 A stable molecule containing atoms of the elements, **A**, **D** and **E** has the following structure.



Which of the following is a possible combination of elements?

	A	D	E
A	N	P	Cl
B	O	S	Cl
C	B	C	H
D	P	N	F

- 7 The graph of P / kPa against $(1 / V) / \text{m}^{-3}$ was obtained for 0.200 g of an unknown gas at 25°C .



What is the relative molecular mass of the gas?

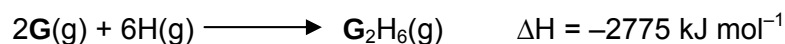
- A 17.6
B 21.0
C 23.7
D 42.0
- 8 The Gibbs free energy change of a system determines whether a reaction is spontaneous, while the equilibrium constant indicates the extent of reaction.

What does the following pair of values for a reaction system indicate?

	<i>values</i>
$\Delta G^\ominus_{\text{rxn}}$	-50.8
K_c	2.35×10^{-4}

- A reaction goes to completion
B position of equilibrium lies to the right
C position of equilibrium lies to the left
D no reaction

- 9 The equation below represents the combination of gaseous atoms of non-metal **G** and of hydrogen to form gaseous **G₂H₆** molecules.



The bond energy of a **G–G** bond is +405 kJ mol^{–1}.

What is the bond dissociation energy of a **G–H** bond?

- A +328 kJ mol^{–1}
 B +395 kJ mol^{–1}
 C +540 kJ mol^{–1}
 D +681 kJ mol^{–1}
- 10 Which of the following statements about a zinc–silver cell is true?
- A Silver metal will be deposited on the anode.
 B Electrons flow from the silver(I) oxide electrode to the zinc electrode.
 C Zinc metal will be deposited on the cathode.
 D The e.m.f of the cell will remain constant if the mass of silver(I) oxide is increased.
- 11 The table below contains the reduction potential values for various copper and tin species.

half-equations	E°/ V
$\text{Cu}^{2+}(\text{aq}) + \text{e} \rightleftharpoons \text{Cu}^+(\text{aq})$	+0.15
$\text{Cu}^+(\text{aq}) + \text{e} \rightleftharpoons \text{Cu(s)}$	+0.52
$\text{Sn}^{2+}(\text{aq}) + 2\text{e} \rightleftharpoons \text{Sn(s)}$	–0.14
$\text{Sn}^{4+}(\text{aq}) + 2\text{e} \rightleftharpoons \text{Sn}^{2+}(\text{aq})$	+0.15

Which of the following statements is correct?

- A Both Cu^+ and Sn^{2+} undergo disproportionation.
 B Only Cu^+ undergoes disproportionation.
 C Only Sn^{2+} undergoes disproportionation.
 D Neither Cu^+ nor Sn^{2+} undergoes disproportionation.

- 12 A gas **X** dissociates on heating to set up the equilibrium below.



X was heated at constant pressure p at a certain temperature. The equilibrium partial pressure of **X** was found to be $\frac{1}{7}p$. What is the value of equilibrium constant, K_p at this temperature?

- A $\frac{6}{7} p$
- B $\frac{9}{7} p$
- C $\frac{36}{7} p$
- D $6 p$
- 13 Given that the K_{sp} for silver hydroxide, AgOH, is $1.52 \times 10^{-8} \text{ mol}^2 \text{ dm}^{-6}$ at 298 K, what is the pH of the solution at equilibrium?
- A 2.6
- B 3.9
- C 10.1
- D 11.4
- 14 The dissociation constant, K_w , for the ionisation of water, $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$, at different temperatures is given below.

temperature / °C	$K_w / \text{mol}^2 \text{ dm}^{-6}$
0	1.15×10^{-15}
25	1.00×10^{-14}
50	5.50×10^{-14}

What can be deduced from this information?

- A $[\text{H}^+]$ and $[\text{OH}^-]$ are equal only at 25°C.
- B The equilibrium lies furthest to the right at 0 °C.
- C The forward reaction is exothermic.
- D The pH of the water decreases as temperature increases.

- 15 The rate of removal of the pain-killing drug paracetamol from the body is a first order reaction with a rate constant, $k = 0.26 \text{ hr}^{-1}$.

How long will it take to remove 75% of the paracetamol that a patient consumes?

- A 2.3 hr
 - B 2.7 hr
 - C 5.3 hr
 - D 8.0 hr
- 16 Which physical property of Period 3 elements from sodium to chlorine shows a general decrease when moving across the period?
- A ionic radius
 - B first ionisation energy
 - C maximum oxidation number in oxide
 - D electrical conductivity
- 17 *Use of the Data Booklet is relevant to this question.*

A 5.00 g sample of an anhydrous Group II metal nitrate loses 3.29 g in mass when heated strongly.

Which metal is present?

- A magnesium
 - B calcium
 - C strontium
 - D barium
- 18 Which one of the following properties describes the trend for the hydrogen halides as stated below?



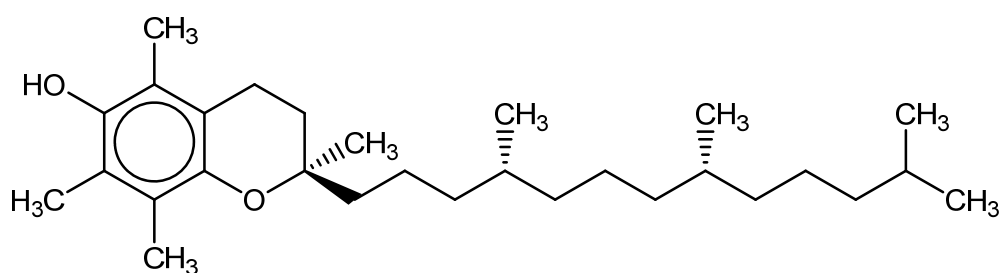
- A thermal stability of the compounds
- B boiling point of the compounds
- C ease of oxidation of the compounds
- D acidity of the solutions

- 19 Platinum(IV) chloride combines with ammonia to form compounds in which the coordination number of platinum is 6. A formula unit of one of the compounds contains a cation and only one chloride ion.

What is the formula of this compound?

- A $\text{Pt}(\text{NH}_3)_6\text{Cl}_4$
- B $\text{Pt}(\text{NH}_3)_5\text{Cl}_4$
- C $\text{Pt}(\text{NH}_3)_4\text{Cl}_4$
- D $\text{Pt}(\text{NH}_3)_3\text{Cl}_4$

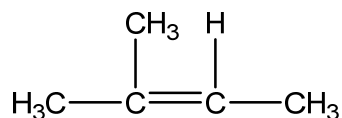
- 20 α -tocopherol is the most famous vitamin E which has the ability to help prevent free radical damage.



α -tocopherol

How many stereoisomers of α -tocopherol are present?

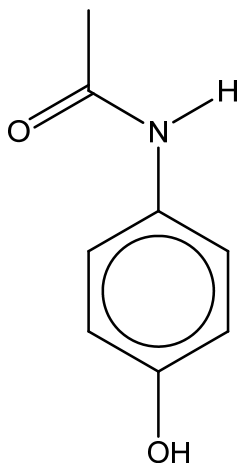
- A 4
 - B 3
 - C 8
 - D 16
- 21 When bromine reacts with 2-methylbut-2-ene in the presence of concentrated aqueous sodium chloride, which of the following is the major product?



2-methylbut-2-ene

- A $(\text{CH}_3)_2\text{CBrCH}(\text{CH}_3)\text{Cl}$
- B $(\text{CH}_3)_2\text{CClCH}(\text{CH}_3)\text{Br}$
- C $(\text{CH}_3)_2\text{CClCH}(\text{CH}_3)\text{Cl}$
- D $(\text{CH}_3)_2\text{CBrCH}(\text{CH}_3)\text{Br}$

- 22 Paracetamol is a common analgesic used in the treatment of fever and headaches.

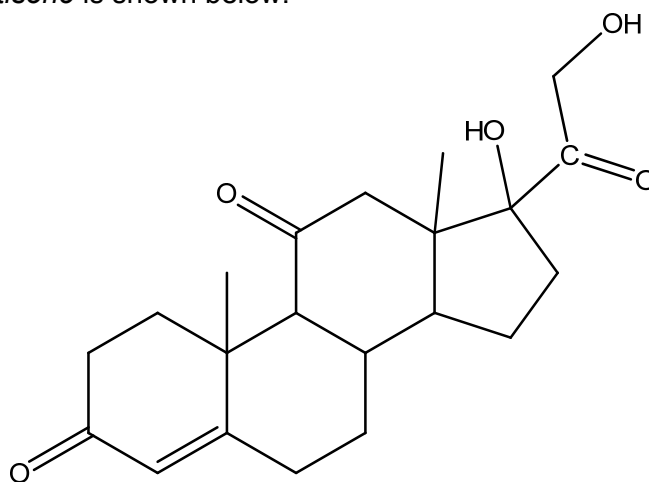


paracetamol

Which deduction about paracetamol can be made from this structure?

- A It will dissolve in aqueous HCl when heated.
- B It will not react with aqueous NaOH.
- C It is a planar molecule.
- D It reacts with Tollen's reagent.

- 23 The structure of *cortisone* is shown below.



cortisone

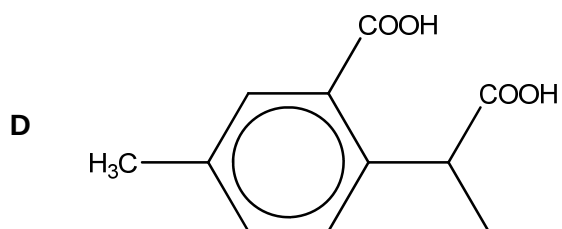
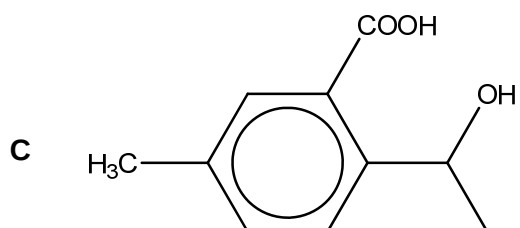
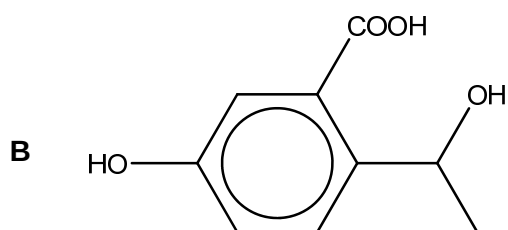
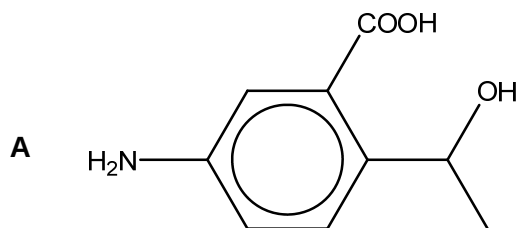
Which of the following statements about the reactions of *cortisone* is likely to be **correct**?

- A 1 mol of cortisone reacts with Na to give 1 mol of H₂ gas.
- B 1 mol of cortisone reacts with sodium carbonate to release 1 mol of CO₂.
- C It gives a yellow precipitate with alkaline aqueous iodine.
- D It reacts with Brady's reagent to form a black precipitate.

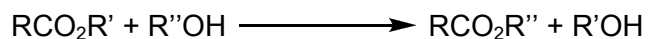
24 Compound **H** has the following properties:

- (i) It does **not** decolourise aqueous bromine;
- (ii) It produces an effervescence when reacted with $\text{Na}_2\text{CO}_3(\text{aq})$ and
- (iii) It gives yellow precipitate with alkaline aqueous iodine.

Which of the following can be **H**?

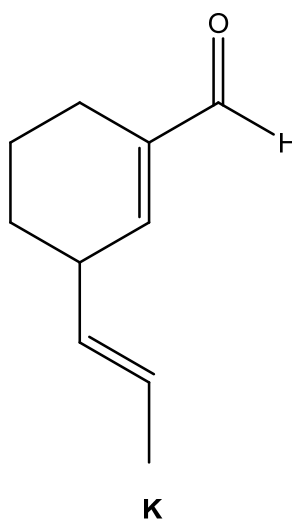
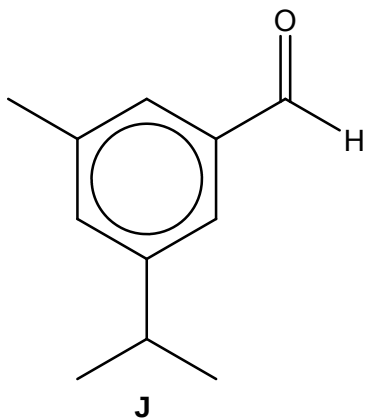


- 25 Transesterification is the process of exchanging the organic group R' of an ester with the organic group R'' of an alcohol.



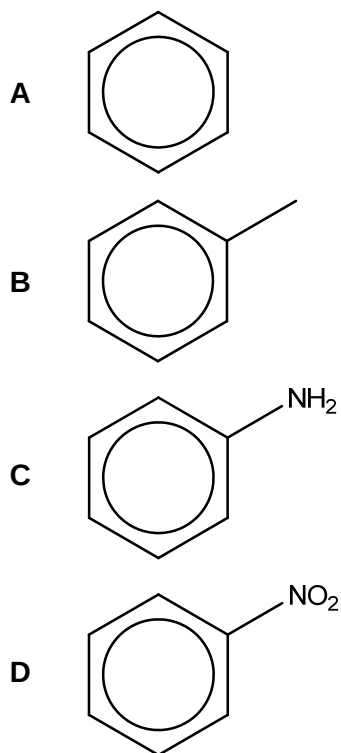
What type of reaction takes place in transesterification?

- A nucleophilic addition
 - B nucleophilic substitution
 - C electrophilic addition
 - D electrophilic substitution
- 26 Which reagents can be used to distinguish between the compounds **J** and **K**?

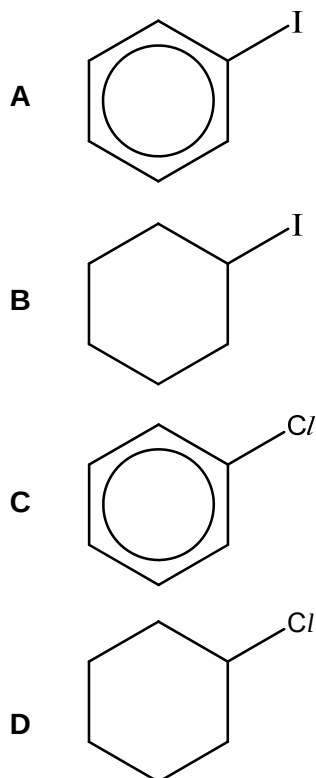


- A $[\text{Ag}(\text{NH}_3)_2]^+$
- B 2,4-dinitrophenylhydrazine
- C Cu^{2+} in $\text{NaOH}(\text{aq})$
- D sodium carbonate

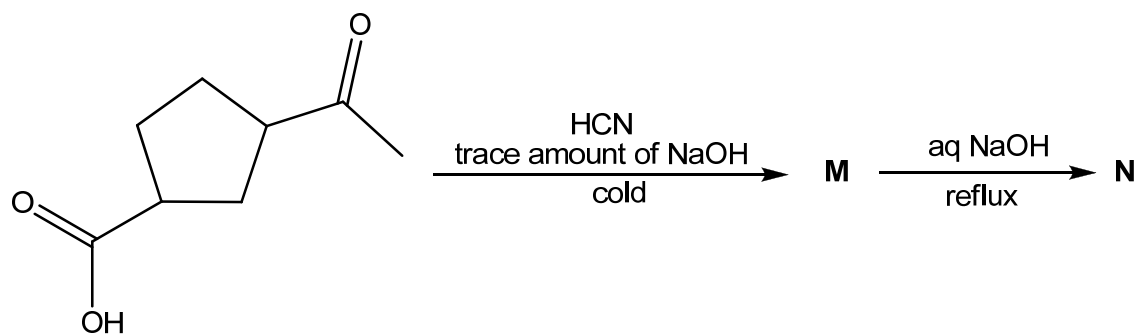
27 Which of the following compounds would undergo electrophilic substitution most readily?



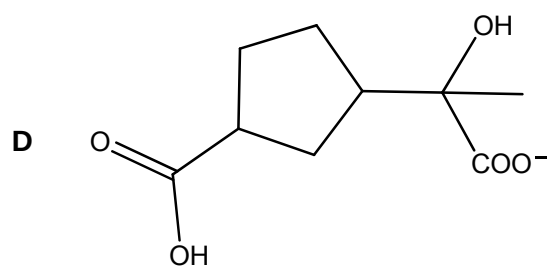
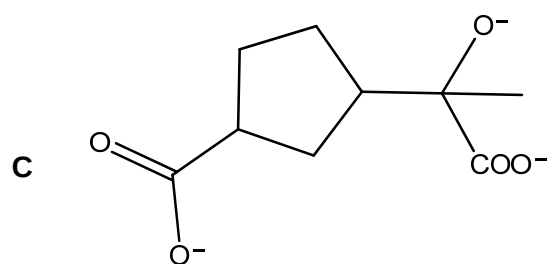
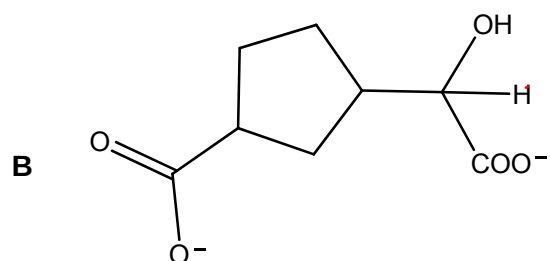
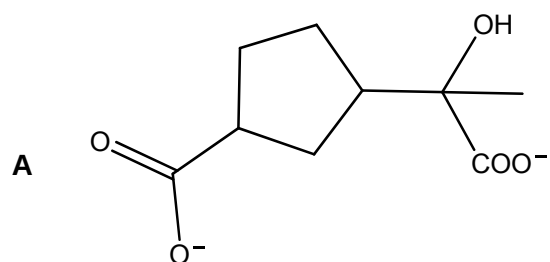
28 A halogen derivative, L, was first warmed with aqueous sodium hydroxide, followed by adding excess of dilute nitric acid and aqueous silver nitrate. A precipitate was produced. Dilute aqueous ammonia was then added and a colourless solution is obtained. Which of the following could be the identity of compound L?



29 A reaction sequence is shown below.



Which of the following structures represents the product **N**?



- 30 What stabilises the β -pleated sheet of proteins?
- A hydrogen bonds only
 - B van der Waals' forces only
 - C electrostatic forces of interactions only
 - D peptide bonds

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

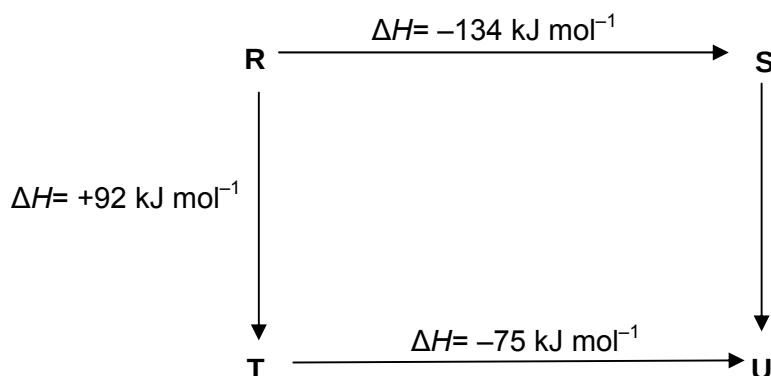
No other combination of statements is used as a correct response.

- 31** Gaseous particle **P** has a proton number of $n+1$ and a charge of +2. Gaseous particle **Q** has proton number n and is isoelectronic with **P**.

Which of the following statements are true?

- 1** **Q** has a larger radius than **P**.
- 2** **Q** requires less energy than **P** when an electron is removed from each particle.
- 3** **Q** releases less energy than **P** when an electron is added to each particle.

- 32** The diagram illustrates the energy changes of a set of reactions.



Which of the following statements are correct?

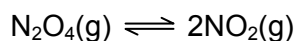
- 1** The enthalpy change for the transformation **R** $\longrightarrow$ **U** is endothermic.
- 2** The enthalpy change for the transformation **S** $\longrightarrow$ **T** is +226 kJ mol⁻¹.
- 3** The enthalpy change for the transformation **S** $\longrightarrow$ **U** is +151 kJ mol⁻¹.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

33 For the reaction:



the numerical values of K_c at two different temperatures are given as follows:

temperature/ K	K_c
x	1.53
298	4.64×10^{-3}

Which of the following statement(s) is / are correct?

- 1** The proportion of NO_2 in the equilibrium mixture is increased by decreasing the pressure.
- 2** The value of **x** is higher than 298K.
- 3** The units for K_c are $\text{mol}^2 \text{dm}^{-6}$.

34 Which of the following could act as buffer solutions?

- 1** NaHCO_3 and Na_2CO_3
- 2** $\text{CH}_3\text{CO}_2\text{H}$ and NaCl
- 3** HNO_3 and NaNO_3

35 An accident on the expressway was reported as follows:

“Drums of bromine broke open after a vehicle crash on the expressway. Traffic was diverted as purple bromine vapour drifted over the road as it is denser than air, causing irritation to drivers’ eyes. Firemen sprayed water over the scene of the accident, dissolving the bromine and washing it away.”

What is **wrong** with the report?

- 1** Bromine vapour is not purple.
- 2** Bromine vapour is less dense than air.
- 3** Bromine does not dissolve in water.

- 36** Four bottles in a school laboratory had missing labels. It was realised that the contents on one of them could easily be identified because on addition of water it would not give fumes of hydrogen chloride.

Water was added to each of the four substances separately. In which substance(s) will there be white fumes observed?

- 1 MgCl_2
- 2 Al_2Cl_6
- 3 SiCl_4

- 37** Which of the following may affect the rate constant for a reaction?

- 1 change in temperature
- 2 change in catalyst
- 3 change in reactant concentration

- 38** Vanadium has the electronic structure $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$. Which of the following vanadium compounds is/are likely to exist?

- 1 V_2O_5
- 2 VOCl
- 3 $\text{K}_2\text{V}_2\text{O}_7$

- 39** Which sequence shows the nitrogen compounds in increasing order of $\text{p}K_b$ value?

- 1 $\text{CH}_3\text{CH}_2\text{NHCH}_3$, $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$, CH_3CONH_2
- 2 $\text{C}_6\text{H}_5\text{NH}_2$, NH_3 , CH_3NH_2
- 3 $\text{CH}_3\text{CH}_2\text{N}(\text{CH}_3)_2$, $\text{C}_6\text{H}_5\text{NH}_2$, $\text{CH}_3\text{CH}_2\text{NHCH}_3$

- 40** Samples of the gases CH_3Cl and Cl_2 are mixed together and irradiated with light. Which statement(s) is/are true?

- 1 Chlorine atoms are formed in the initiation step.
- 2 The chlorine radical acts as a catalyst in the propagation steps.
- 3 Ethane is produced in trace amounts in the termination step.

